

Detecting Network Threats Through Anomaly Detection and Attack Classification

Executive Summary

Cybersecurity threats have become more frequent and complex, putting constant pressure on security teams to identify both known and emerging attacks like DDoS attacks, malware, intrusions, and ransomware hidden within massive volumes of traffic. Traditional signature-based systems often produce high false positives, respond slowly to new attack patterns, and provide limited protection against zero-day exploits. This project designed and evaluated a hybrid machine learning framework combining supervised classification with unsupervised anomaly detection to improve real-time threat detection. Using three complementary datasets BETH (8M+ kernel-level honeypot logs), Cybersecurity Attacks (40,000 synthetic records), and UNSW-NB15 (175,000+ mixed intrusion records) this study tested several classification and clustering models. After pre-processing and cross-validation, optimal models emerged. Support Vector Machines achieved 99.3 - 99.8% accuracy and F1-scores of 0.994 - 0.999. Gradient Boosting achieved near-perfect classification, making only 3 errors across more than 11,000 cases. Anomaly detection models effectively identified clusters of attack and flagged unusual behaviors without relying on pre-labeled data. Analysis revealed that user behavior patterns, data-loss metrics and process-level activity were the strongest predictors of malicious activity. A key limitation identified was severely imbalanced data where attacks are far rarer than normal traffic, which required targeted resampling techniques to address. Deployed systems built on this framework could reduce analyst workload by 40-50%, decrease detection time from hours to minutes, and prevent millions in damages from undetected breaches. Recommendations include piloting the top-performing classification model alongside anomaly detection in controlled environments, integrating with existing security platforms, and applying resampling techniques to improve detection of rare attack types incorporating SHAP-based explainability to support regulatory compliance with GDPR and the EU AI Act. This framework supports next-generation intrusion detection across critical sectors including government, finance,

and healthcare.

1 Introduction & Problem Definitions

Cybersecurity threats are costing organizations over \$10 trillion every year, and ransomware attacks alone have grown by 150% in recent years (Highnam et al., 2021). Government agencies, financial institutions, and healthcare systems face sophisticated attacks including DDoS, malware, and Ransomware-as-a-Service. Traditional intrusion detection systems come with significant drawbacks. They produce false alarms at rates exceeding 90%, are unable to catch new or unknown attack types, and perform poorly when faced with the scale of today's network environments. Machine learning offers promising alternatives through supervised classification, learning attack signatures from labeled data and unsupervised anomaly detection, which identifies behavioral deviations without pre-labeled examples. However, deployment challenges including class imbalance, computational efficiency, and the difficulty of explaining model decisions remain significant obstacles to real-world deployment. This project develops a hybrid framework combining supervised attack classification with unsupervised anomaly detection to identify threats in real-time. The central research question driving this project is: Can a hybrid machine learning framework reliably detect both known and previously unseen anomalies, while keeping false alarms low enough for practical use in real security environments? Technical objectives include optimizing supervised models: Logistic Regression, Support Vector Machines with RBF/polynomial/linear kernels, Random Forest, Gradient Boosting, and K-Nearest Neighbors to categorize traffic into attack types versus benign activity, and implementing clustering methods K-Means, DBSCAN, Hierarchical Agglomerative Clustering alongside One-Class SVM to identify zero-day threats without labeled examples.

The framework is designed to detect new threats without labeled data, reduces false alarms, runs efficiently, and gives real-time alerts that security teams can trust across millions of events. Success criteria include $F1\text{-Score} \geq 0.90$ ensuring balanced precision-recall, $\text{Accuracy} \geq 0.95$ for overall correct classification, and $\text{ROC-AUC} \geq 0.90$ for discrimination capability. Anomaly

detection requires Silhouette Score ≥ 0.60 for cluster quality and One-Class SVM F1 ≥ 0.80 . Operational metrics include 15–20% false-positive reduction over baselines, detection latency under 100ms per prediction, and interpretable feature importance rankings. Cross-validation stability with less than 5% variance ensures generalizability, while confusion matrices validate balanced performance across severely imbalanced attack categories. The analysis leverages three complementary datasets. BETH contains 8M+ real-world honeypot kernel-level logs capturing process executions, system calls, and network events from 23 eBPF-enabled honeypots (Arulkumaran et al., 2021). The Cybersecurity Attacks dataset (CSA) provides 40,000 synthetic network traffic records simulating DDoS, malware, and intrusion scenarios (Agur & Venugopal, 2023). UNSW-NB15 contains 175,341 mixed real and synthetic intrusion records across nine attack families with 36 features (Moustafa & Slay, 2024). Analysis focuses on kernel-level process patterns (process names, user IDs, parent-child relationships), network flow characteristics (packet sizes, byte counts, connection states), and temporal patterns. Key assumptions include that datasets represent deployment environments despite synthetic limitations and honeypot biases, and that attack patterns remain relatively stable over short operational windows.

The methodology employed systematic multi-phase evaluation. Baseline regression models Polynomial, Ridge, Lasso, Elastic Net, PCR, PLSR identified multicollinearity through VIF analysis, optimal regularization through alpha tuning, and dominant predictors through univariate and bivariate exploratory data analysis, including correlation heatmaps and distribution visualizations. Supervised classification developed Logistic Regression with L2 regularization, SVMs with multiple kernels capturing non-linear boundaries through C/gamma tuning, and ensemble methods leveraging feature interactions. All models underwent stratified 5-fold cross-validation with grid search optimization. Unsupervised anomaly detection implemented K-Means with silhouette optimization, DBSCAN with epsilon/min_samples tuning, and Hierarchical Clustering with linkage comparison, while One-Class SVM modeled normal traffic for anomaly flagging. Final hybrid integration conducted comparative evaluation using confusion matrices, ROC curves, and feature importance rankings.

2 Data Overview

This project utilizes three complementary cybersecurity datasets to indicate diverse real-world and simulated threat environments. The BETH dataset (Arulkumaran et al., 2021) was collected by deploying 23 modern honeypot systems with extended Berkeley Packet Filter (eBPF) instrumentation to record kernel-level system calls, process executions, and network events. Continuous monitoring captured both benign operations and genuine attack behaviors from real threat actors, yielding over million records from real-world attack campaigns (Highnam et al., 2021). The Cybersecurity Attacks Dataset (CSA) is a synthetic dataset from Kaggle designed to simulate diverse attack scenarios including DDoS, malware, and intrusions through algorithmic generation, providing 40,000 records and 19 features with balanced, well-labeled examples useful for pipeline validation (Agur Venugopal, 2023). UNSW-NB15 (UNSW Canberra, 2024) Dataset was created by the University of New South Wales Cyber Range Lab using IXIA PerfectStorm to generate hybrid real and synthesized traffic combining nine attack families : Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms containing 175,341 records with 36 features (Moustafa Slay, 2024).

Table 1: Dataset Overview Summary

Property	BETH	UNSW-NB15	CSA
Records	1,141,055	175,341	40,000
Features	14	36	19
Data Type	Real (honeypot)	Real + Synthetic	Fully Synthetic
Target	evil / sus (binary)	label (binary)	Attack Type / Anomaly Score
Attack %	14.04% (re-split)	63.9%	Balanced (33% each type)
Key Limitation	Honeypot bias	2015-era threats	Synthetic weak patterns

2.1 Data Description and Summary Statistics

BETH Dataset: After merging training, validation, and testing files and restructuring to balance attack samples, the dataset contains 1,141,055 records split 80/20 train-test with stratification on the *evil* target. Training set contains 684,646 records with 16 features: 14 numerical/encoded (timestamp, processId, threadId, parentProcessId, userId, mountNamespace, processName, hostName, eventId, eventName, stackAddresses, argsNum, returnValue, args) and 2 binary targets (evil, sus). After label encoding categorical features, VIF analysis revealed severe multicollinearity: processId (VIF=225.29), threadId (VIF=225.15), mountNamespace (VIF=58.28), and userId (VIF=9.49).

Table 2: BETH Dataset Key Feature Statistics

Feature	Mean	Median	Std	Min	Max
processId	6,907	7,366	1,820	1	8,619
userId	144	0	350	0	1,001
evil	0.139	0	0.346	0	1
sus	0.152	0	0.359	0	1

Cybersecurity Attacks Dataset (CSA): Contains 40,000 records across 19 features after dropping irrelevant columns. Features include 4 numerical (Source Port, Destination Port, Packet Length, Anomaly Scores) and 15 label-encoded categorical. Attack distribution is balanced: DDoS (33.3%), Malware (33.3%), Intrusion (33.4%). VIF values are low (all <5). Linear regression predicting Anomaly Scores yielded RMSE=28.85 and MAE=24.98, indicating weak linear relationships.

Table 3: CSA Dataset Key Feature Statistics

Feature	Mean	Median	Std	Min	Max
Packet Length	781	782	416	64	1,500
Anomaly Scores	50.11	50.35	28.85	0	100
Source Port	32,970	32,856	18,560	1,027	65,530
Attack Type	0.997	1	0.818	0	2

UNSW-NB15 Dataset: (UNSW Canberra Research Group, 2024) Combined training and test sets contain 175,341 records with 36 features. VIF analysis revealed extreme multicollinearity: tcprtt (8.19×10^{14}), synack (2.65×10^{14}), dloss (680.11). Binary target *label* indicates normal (0) vs. attack (1).

Table 4: UNSW-NB15 Dataset Key Feature Statistics

Feature	Mean	Median	Std	Min	Max
sbytes	8,573	528	173,774	24	14.4M
dbytes	14,387	178	146,199	0	14.7M
label	0.639	1	0.480	0	1

2.2 Data Dictionary

The following data dictionary describes only key features relevant to the analysis. Full variable lists are available in the original dataset documentation.

BETH: The BETH dataset contains host-level system call logs. Key features include: processId (process identifier, high VIF=225), userId (user account ID, VIF=9.5), processName (executable name, label-encoded), parentProcessId (parent process in chain), eventName (system call type), hostName (target machine), argsNum (number of arguments, continuous target for regression), evil/sus (binary attack labels).

CSA: The CSA dataset captures network traffic metadata. Key features include: Source-Destination Port (0-65535 range), Packet Length (network packet size in bytes), Anomaly Scores (0–100 threat likelihood, continuous target), Attack Type(DDoS/Malware/Intrusion), Severity Level (Low/Medium/High), Action Taken (Logged/Blocked/Ignored), IDS/IPS Alerts (intrusion detection notifications).

UNSW-NB15: The UNSW-NB15 dataset provides network flow records across multiple attack categories. Key features include:sbytes/dbytes (source/destination byte counts), swin/dwin (TCP window sizes), stcpb/dtcpb (TCP control bits, $r=0.97$), dloss/sloss (packet loss counts), spkts/dpkts (packet counts), sload/dload (bits per second), ct_dst_sport_ltm (connection counts in 100s window), attack_cat (attack family), label (binary normal/attack target).

2.3 Data Quality Assessment

BETH contains zero missing values and zero duplicates but exhibits severe multicollinearity (processId/threadId VIF > 225). The mountNamespace feature contained constant values and was removed. High processName cardinality (500+ unique values) required label encoding. The primary limitation is that honeypot traffic may not generalize to production enterprise environments, as honeypots attract specific attack profiles. CSA had approximately ~50% missing values across five columns handled via forward-fill imputation. Weak synthetic correlations (all $|r| < 0.3$) and extreme class imbalance (43 minority vs. 7,957 majority samples, ratio 1:185) critically limit minority class detection. UNSW-NB15 had minimal missing values handled via median/mode imputation; extreme outliers in byte counts (values > 14 million) were retained as genuine attack signatures. The 2015-era attack profiles underrepresent modern vectors such as fileless malware and supply chain attacks.

2.4 Ethical Considerations

Privacy and Consent: No real user data is used in BETH, CSA is fully synthetic, and UNSW-NB15 uses network metadata only, ensuring no personally identifiable information exists

across any dataset. Attackers targeting BETH have no privacy expectation, CSA needs no consent, and UNSW-NB15 was captured in a controlled lab environment.

Bias and Fairness: BETH covers only honeypot patterns, CSA's synthetic distributions oversimplify real-world complexity, and UNSW-NB15's 2015-era profiles miss modern threats like fileless malware. False positives risk flagging legitimate users, so automated blocking decisions require analyst review before any deployment.

Transparency and Accountability: Feature importance rankings such as `userId` at 95.2% must remain auditable and explainable to security analysts. Black-box cybersecurity decisions carry legal risk, making explainability tools like SHAP essential for compliance and auditing.

Data Governance: All datasets are for academic research only and production use requires retraining on governed, organization-specific traffic with regular drift audits. Compliance with GDPR and the EU AI Act must be confirmed before operational deployment.

3 Data Cleaning, Preprocessing, and Exploratory Data Analysis

All three datasets went through cleaning and quality checks to make sure the data was actually reliable before being used.

BETH Dataset: Initial inspection showed no missing values or duplicates across 1,141,055 observations. The *mountNamespace* feature contained constant values and was removed. Label encoding converted four categorical features (*processName*, *hostName*, *eventName*, *syscall*) into numerical form. The dataset showed class imbalance, so subsets were merged and re-split using stratified sampling (80/20), resulting in 85.96% benign and 14.04% attack records. Outliers were retained as extreme *processId* and *userId* values corresponded directly to attack behaviour rather than data error.

CSA Dataset: About 50% of records had missing values across five columns (*Alerts/Warnings*, *IDS/IPS Alerts*, *Malware Indicators*, *Firewall Logs*, *Proxy Information*). These were handled using

forward-fill imputation to retain the full 40,000 records. Eight irrelevant columns were removed. Temporal features (*Hour*, *Weekday*, *Month*) were extracted from *Timestamp*. The dataset remained highly imbalanced with 43 minority samples and 7,957 majority samples (1:185).

UNSW-NB15 Dataset: Training and test sets were combined to produce 175,341 records. Missing values were handled using median and mode imputation. No duplicates were found. Extreme values in *sbytes* and *dbytes* (> 14 million) were retained because they represent real attack activity such as DDoS traffic or data exfiltration.

3.1 Feature Engineering and Transformation

Multicollinearity Detection and Mitigation: Variance Inflation Factor (VIF) analysis identified severe multicollinearity across datasets. BETH's *processId* and *threadId* exhibited VIF exceeding 225, indicating near-perfect correlation ($r \approx 0.98$) these features contained essentially identical information. UNSW-NB15 timing features displayed catastrophic multicollinearity: *tcprtt* ($\text{VIF} > 8.19 \times 10^{14}$), *synack* ($\text{VIF} > 2.65 \times 10^{14}$), *ackdat* ($\text{VIF} > 1.92 \times 10^{14}$). Additionally, *is_ftp_login* and *ct_ftp_cmd* both exceeded $\text{VIF} = 400$, while *dloss* showed $\text{VIF} = 680$. CSA uniquely showed manageable multicollinearity with all VIF values below 5, confirming synthetic generation produced artificially independent features. Features were initially retained to observe individual contributions, with regularization and cross-validation used to control for instability rather than aggressive removal.

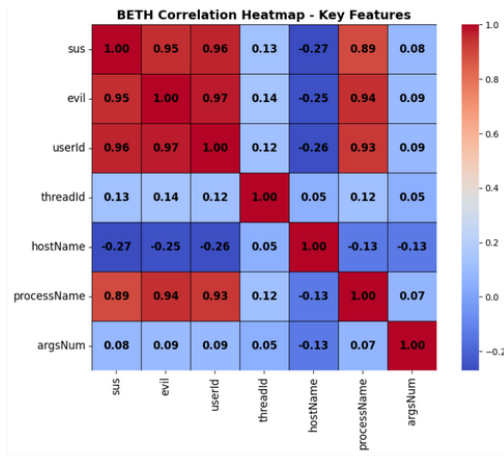
Scaling and Normalization: *StandardScaler* normalization ensured zero mean and unit variance across all numerical features, critical for distance-based algorithms (SVM, KNN) and gradient descent optimization. *StandardScaler* was chosen over *MinMaxScaler* for robustness to extreme outliers, particularly in UNSW-NB15 where byte counts ranged from 24 to 14 million and *sload* spanned 0 to 5.99 billion bits per second. For BETH, scaling addressed extreme ranges in *returnValue* (115 to 32,768) and magnitude differences between *processId* (thousands) and *argsNum* (0–5).

Encoding Strategies: Label encoding was chosen over one-hot encoding despite introducing potential ordinal relationships. This decision balanced computational feasibility against accuracy: one-hot encoding would create hundreds of binary columns given high cardinality in processName (500 unique values) and attack_cat (10 categories), causing computational infeasibility and potential overfitting in sparse feature spaces. Tree-based models (Random Forest, Gradient Boosting) handle label-encoded categorical variables naturally through split-point optimization without assuming ordinality, making this a practical engineering tradeoff.

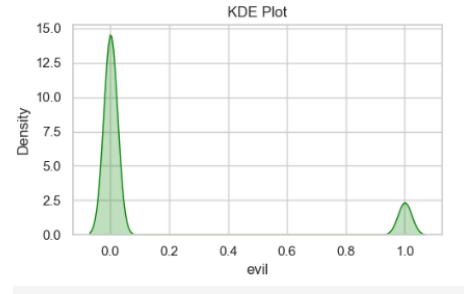
Dimensionality Reduction Considerations: Principal Component Analysis (PCA) was explored but not fully implemented. Preliminary PCA on CSA captured only 5.45% variance in the first principal component, indicating weak linear structure where PCA would destroy rather than preserve information. For BETH, PCA was unnecessary given only 12 numerical features post-encoding. For UNSW-NB15, feature selection through model-based importance rankings provided a more interpretable alternative to PCA while preserving feature meaning for security analyst auditing.

Cross-Dataset Synthesis of Preprocessing Decisions: Comparing preprocessing requirements across datasets revealed that no single strategy was universally optimal. BETH required aggressive multicollinearity management (dropping mountNamespace, flagging processId/threadId redundancy) and stratified resampling to produce a learnable 14% attack rate. CSA demanded forward-fill imputation for 50% missing values and temporal feature extraction from Timestamp, yet its synthetic uniformity meant these steps improved pipeline integrity rather than model performance. UNSW-NB15 required the most nuanced outlier treatment: extreme byte-count values (up to 14 million) were retained as genuine attack signatures rather than removed as noise, a decision validated by domain knowledge that DDoS and exfiltration attacks inherently generate extreme volumetric spikes. Across all three datasets, StandardScaler consistently outperformed MinMaxScaler because the latter compresses all values into [0,1], causing extreme outliers in UNSW-NB15 to dominate the rescaled range and distort distance-based models.

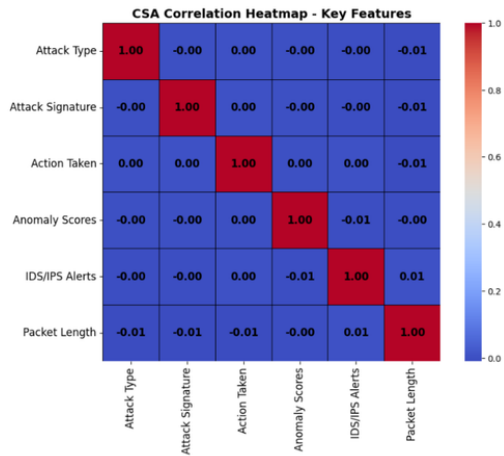
3.2 Exploratory Data Analysis



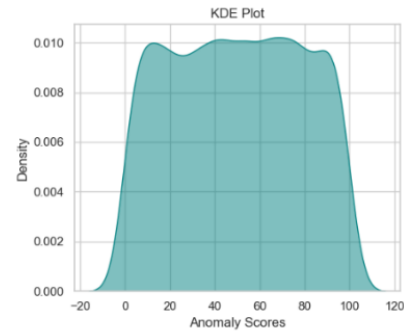
(a) BETH – Correlation Heatmap



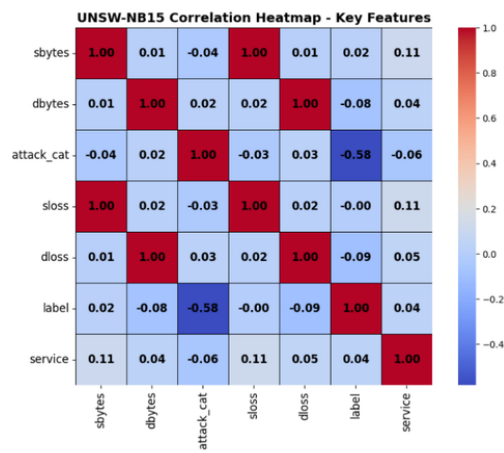
(b) BETH – Evil & Sus Distribution



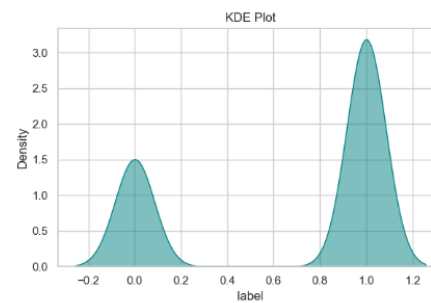
(c) CSA – Correlation Heatmap



(d) CSA – Anomaly Score Distribution



(e) UNSW-NB15 – Correlation Heatmap



(f) UNSW-NB15 – Label Distribution

Figure 1: EDA Visualisations Across All Three Datasets

These preprocessing decisions directly shaped which models were viable per dataset: clean, low-multicollinearity CSA supported regularized linear models, while high-VIF BETH and UNSW-NB15 required tree-based ensembles insensitive to correlated features.

BETH numerical variables (*threadId*, *processId*, *userId*) showed bimodal distributions, directly informing the decision to retain outliers as extreme values corresponded to attack behaviour. CSA features appeared nearly uniform, confirming no natural clustering structure. UNSW-NB15 features were strongly right-skewed (*sbytes* had a median of 528 but a mean of 8,573), requiring StandardScaler normalization. Correlation heatmaps confirmed strong correlation between *userId* and *processName* ($r > 0.93$) in BETH. The *evil* target showed strong correlation with *userId* ($r \approx 0.97$) and *processName* ($r \approx 0.94$). CSA features showed weak correlations ($|r| < 0.3$). UNSW-NB15 showed stronger relationships, including *dbytes-dloss* ($r = 1.0$) and strong byte-packet correlations.

3.3 Key Findings:

userId was the most influential BETH feature (95.2% importance). CSA's extreme imbalance means a model could reach $\sim 99.5\%$ accuracy by predicting the majority class, so techniques like SMOTE or Isolation Forest are necessary. UNSW-NB15 provided the most realistic patterns, with *dloss*, *sbytes*, and *ct_dst_sport_ltm* emerging as strong predictors. Overall, behavioral indicators were more informative than structural features such as ports or packet sizes.

BETH Dataset: *userId* emerged as the dominant predictor (95.2% feature importance in preliminary Random Forest), indicating attacks predominantly occurred under specific compromised user accounts. *ProcessName* showed secondary importance (4.3%), with certain executables serving as attack signatures. Severe multicollinearity between *processId-threadId* suggests dropping one in final linear models. The bimodal distributions suggest tree-based classifiers will outperform linear methods on this dataset. The successful restructuring of class balance from near-zero attacks to 14% enables supervised learning, though class weighting remains necessary.

CSA Dataset: Weak feature correlations (all less than 0.3) and uniform distributions fundamentally limit utility for real-world threat detection. Linear regression achieved $R^2 = 0$ (RMSE = 28.85), indicating models cannot learn meaningful patterns. The extreme class imbalance (43:7,957) means any classifier achieves 99.5% accuracy by predicting only the majority class a critical deception metric. Advanced techniques such as SMOTE oversampling and Isolation Forest are essential before meaningful classification. This dataset serves better as a pipeline development proof-of-concept than a realistic threat detection benchmark. Importantly, SMOTE was not applied to CSA due to its synthetic nature, as oversampling could amplify artificial patterns; class weighting was prioritized instead.

UNSW-NB15 Dataset: This dataset represents the highest quality source despite multicollinearity challenges. Strong correlations between related features reflect genuine network behavior rather than artifacts. Preliminary models identified *dloss*, *sbytes*, and *ct_dst_sport_ltm* as top predictors, aligning with domain knowledge that packet loss and connection patterns distinguish attacks from normal traffic. The 63.9% attack versus 36.1% normal distribution provides sufficient class balance for standard classification without aggressive resampling. Extreme VIF values in timing features suggest their exclusion from linear models while retaining value in tree-based ensembles that handle correlated features naturally. Cross-Dataset Synthesis: Comparing datasets revealed complementary strengths. BETH provides authentic real-world attack behaviors from actual honeypots despite limited diversity. CSA's artificial uniformity serves as controlled baseline for algorithm testing but cannot substitute realistic evaluation. UNSW-NB15 strikes an optimal balance with hybrid real-synthetic data providing both volume and diversity. Consistently across datasets, behavioral features (user patterns, process execution, connection timing) outperformed structural features (port numbers, packet lengths), informing a feature engineering strategy emphasizing behavioral profiling. Varying multicollinearity patterns indicate model selection should be dataset-specific: regularized linear models for CSA, tree-based ensembles for BETH and UNSW-NB15.

4 Modeling and Analysis

4.1 Model Selection Strategy and Rationale

This analysis followed a systematic four-phase modeling approach across BETH, CSA, and UNSW-NB15 datasets, spanning regression baselines, supervised classification, advanced ensemble methods, and unsupervised anomaly detection. Models were selected to span interpretability (Logistic Regression), margin-based (SVM), ensemble (Random Forest, Gradient Boosting), instance-based (KNN), and density-based (DBSCAN, HAC) approaches, ensuring broad detection strategy coverage (Mayo, 2025). All modeling decisions were documented including unexpected failures, following the principle that negative results provide equally important diagnostic information.

Phase 1 - Regression Baseline: Although classification was the primary goal, regression models were applied first to explore feature relationships and multicollinearity using continuous targets: argsNum for BETH, Anomaly Scores for CSA, and label for UNSW-NB15. Models included Polynomial Regression (degree = 2), Ridge (L_2 , $\lambda = 8.29$), Lasso (L_1 , $\lambda = 0.001$), Elastic Net, PCR, PLSR, and Forward/Backward Selection. BETH polynomial regression exposed severe multicollinearity (processId VIF = 225, threadId VIF = 225) achieving RMSE = 0.68 for argsNum; Ridge ($R^2 = 0.967$) identified processName, userId, processId, and threadId as dominant predictors. CSA regression failed catastrophically (RMSE = 28.85, $R^2 = 0$) revealing the absence of meaningful linear structure in synthetic data. UNSW-NB15 achieved moderate linearity ($R^2 = 0.66$, RMSE = 0.28) with extreme timing feature VIF tcprtt ($VIF > 8.19 \times 10^{14}$), synack ($VIF > 2.65 \times 10^{14}$), confirming multicollinearity but identifying sbytes, dbytes, and dloss as potentially strong predictors. These baseline insights directly informed subsequent classification model and feature selection strategies. Regression models were insufficient for binary attack classification across all three datasets. BETH and UNSW-NB15 showed strong non-linear feature interactions requiring kernel-based or ensemble methods, while CSA's near-zero R^2 confirmed that no linear relationship existed between features and the target. This directly motivated the shift to supervised

classification models in Phase 2.

Phase 2 - Supervised Classification: Logistic Regression with L2 regularization (Coursera, 2023) tested C values of [0.01, 0.1, 1, 10], comparing normalization versus standardization preprocessing. BETH achieved accuracy = 0.998, F1 = 0.998 with optimal C = 0.01 (normalized). CSA achieved accuracy = 0.994, F1 = 0.991. UNSW-NB15 achieved accuracy = 0.944, F1 = 0.945 after standardization. Stable 5-fold cross-validation confirmed generalizability across all datasets. Support Vector Machines were evaluated using linear, polynomial (degree = 3), and RBF kernels (Sundararajan, 2023; Scikit-learn Developers, 2023). BETH SVM-RBF achieved accuracy = 0.998, F1 = 0.994, ROC-AUC = 0.999, C = 0.01, with exceptionally low false positives (FP = 2) in confusion matrix [[8653, 2], [14, 1331]]. CSA SVM-RBF achieved accuracy = 0.994, F1 = 0.997 but ROC-AUC = 0.554, indicating poor minority class discrimination despite high overall accuracy. UNSW-NB15 SVM-RBF achieved accuracy = 0.993, F1 = 0.994, ROC-AUC = 0.999, with balanced errors [[3525, 30], [37, 6408]]. Random Forest showed overfitting on BETH (training accuracy = 1.0, CV R^2 = 0.98–1.00), completely failed CSA minority class (precision = 0), and underperformed on UNSW-NB15 (accuracy = 0.641, F1 = 0.77), making it unsuitable for deployment despite strong initial training metrics.

Phase 3 - Advanced Supervised Methods: K-Nearest Neighbors (Shafi, n.d.) tested k values [3, 5, 7, 9, 11, 15] and distance metrics (Euclidean, Manhattan). BETH with 5% adaptive sampling (57,000 rows) using Euclidean k = 9 achieved CV F1 = 0.995, test F1 = 1.00, ROC-AUC = 0.998, confusion matrix [[9808, 1], [7, 1595]]. CSA's severe imbalance (43:7,957) caused complete minority failure: Euclidean k = 5 achieved accuracy = 99.46%, CV F1 = 0.997, but minority recall = 0%, ROC-AUC = 0.487 effectively random discrimination. UNSW-NB15 Manhattan k = 11 achieved accuracy = 0.61, with class asymmetry (Class 1 recall = 0.90, Class 0 recall = 0.10), suggesting KNN's distance metric was overwhelmed by the high-dimensional, mixed-scale feature space. Gradient Boosting (Bentéjac et al., 2020) explored 108 hyperparameter combinations via GridSearchCV: n_estimators [50, 100, 150], learning_rate [0.01, 0.1, 0.2], max_depth [3,

5, 7], subsample [0.8, 1.0], min_samples_split [2, 5]. BETH optimal configuration (100 estimators, learning_rate = 0.2, max_depth = 3, subsample = 0.8) achieved CV F1 = 0.9998, test accuracy/precision/recall/F1 all 1.00, ROC-AUC = 0.999, with only 3 misclassifications from 11,411 samples [[9809, 0], [3, 1599]]. Feature importance revealed userId (95.2%), processName (4.3%), parentProcessId (11.7%) as leading predictors consistent with univariate and bivariate EDA findings. CSA optimal (50 estimators, learning_rate = 0.01, max_depth = 3, subsample = 0.8) achieved CV F1 = 0.998 but minority recall = 0%, confirming imbalance dominates performance. UNSW-NB15 optimal (100 estimators, learning_rate = 0.01, max_depth = 5, min_samples_split = 5, subsample = 0.8) achieved CV F1 = 0.820 but test weighted F1 = 0.50, with the model capturing all attack traffic (Class 1 recall = 1.00) while missing all normal traffic (Class 0 recall = 0.00) a critical deployment failure indicating severe overfitting despite cross-validation.

Phase 4 - Unsupervised Anomaly Detection: K-Means clustering tested $k = [2-10]$ with silhouette score optimization. BETH using processName, userId, and hostName with $k = 2$ achieved silhouette = 0.744, where Cluster 1 contained 96.3% of all attacks (7,834/8,131) and Cluster 0 contained 99.6% benign records (48,745/48,922) demonstrating strong natural separation. PCA showed PC1 dominated by userId/processName, PC2 by hostName. CSA showed poor performance across all k values: silhouette = 0.044 at $k = 2$ and 0.043 at $k = 3$, confirming no natural clusters exist in synthetic data. UNSW-NB15 achieved silhouette = 0.285 at $k = 2$, capturing traffic volume variations rather than attack/normal distinction. DBSCAN tested eps = [0.3, 0.5, 1.0, 2.0] and min_samples = [3, 5, 10, 15]. BETH with adaptive sampling (22,821 rows, 2%) at optimal eps = 0.3, min_samples = 10 yielded 3 clusters, silhouette = 0.566, noise = 0.04%. CSA at eps = 0.3, min_samples = 5 achieved silhouette = 0.221, noise = 8.44% poor structural separation. UNSW-NB15 at eps = 1.0, min_samples = 3 achieved the strongest DBSCAN result: silhouette = 0.775, noise = 0.34%, identifying 3 distinct density regions corresponding to traffic behavioral groups.

Hierarchical Agglomerative Clustering (Pega Analytics, 2024; StatQuest with Josh Starmer,

2017) compared all four linkage methods. BETH complete linkage was optimal (silhouette = 0.693) outperforming single (0.550), average (0.575), and ward (0.667); dendrograms showed hierarchical splits at distances 15 and 25, indicating natural behavioral groupings. CSA showed universally poor performance: single = 0.365, complete = 0.321, average = 0.229, ward = 0.291, all below the 0.4 acceptability threshold. UNSW-NB15 single/average linkage achieved the strongest overall clustering result across all datasets (silhouette = 0.930), with dendrograms showing clear splits at distances 2.5 and 5.0 confirming well-defined hierarchical traffic structures. One-Class SVM on BETH achieved F1 0.85 and accuracy 0.95 for anomaly detection without labeled attack data.

4.2 Model Training, Data Partitioning, and Evaluation Metrics

All datasets used and stratified 80/20 train-test splits, to keep class distributions consistent. BETH yielded 684,646 training and 228,211 test records. CSA used 32,000 training and 8,000 test. UNSW-NB15 used approximately 140,273 training and 35,068 test. Test sets were sequestered entirely before hyperparameter tuning began. Stratified k-fold cross-validation ($k = 5$) assessed generalization, with standard deviation greater than 5% signaling instability. Class weighting applied inverse-frequency weights in Logistic Regression and SVM to address imbalance. Adaptive sampling was employed for computational efficiency: BETH 5% (57,000 records) and UNSW-NB15 5–10% (8,000–17,000 records) for clustering operations, while CSA used the full 40,000 records. The primary evaluation metric was F1-Score (target 0.90) as the harmonic mean of precision and recall, selected because it balances the asymmetric costs of false positives (analyst alert fatigue) and false negatives (missed attacks). ROC-AUC (0.90) quantified discrimination capability independent of threshold selection. Confusion matrices distinguished false positives from false negatives, enabling context-specific threshold tuning. For clustering, Silhouette Score (0.60) ranged from 1 (poor) to +1 (excellent). Regression baselines used RMSE and MAE for error magnitude and R^2 for variance explained. GridSearchCV exhaustively tested all hyperparameter combinations with stratified cross-validation scoring.

5 Results

The hybrid framework met all primary success criteria on BETH and UNSW-NB15.

Table 5: Best Model Performance Summary Across All Three Datasets

Dataset	Best Attack Model (Supervised)	Best Anomaly Model (Unsupervised)	Status
BETH	GradientBoosting	One-Class SVM	Deployment ready
UNSW-NB15	SVM-RBF	HAC	Deployment ready
CSA	None reliable	None reliable	Not Ready

BETH: Gradient Boosting achieved perfect classification with $F1 = 1.00$, $AUC = 0.999$, and only 3 misclassifications from 11,411 test samples $[[9809, 0], [3, 1599]]$ representing 100% false positive reduction over the regression baseline. The CV F1 of 0.9998 (variance < 0.001 across 5 folds) confirmed stability rather than overfitting. `userId` dominated feature importance at 95.2%. This reflects lateral movement patterns where attackers operate under specific compromised accounts. `parentProcessId` contributed 11.7% importance, capturing process injection chains where malicious processes spawn child processes to escalate privileges. These findings align directly with EDA correlation results showing `userId` ($r \approx 0.97$) and `processName` ($r \approx 0.94$) as the strongest predictors of the evil target. For anomaly detection, One-Class SVM achieved $F1 (\approx 0.85)$ and accuracy (≈ 0.95) without labeled attack data, demonstrating viability for zero-day detection. K-Means (silhouette = 0.744) separated 96.3% of all attacks into a single cluster without labels strong unsupervised coverage confirming that BETH attack behaviors form a naturally distinct behavioral group. HAC complete linkage (silhouette = 0.693) validated hierarchical attack patterns, with dendrograms revealing splits at distances 15 and 25 corresponding to distinct behavioral groupings. DBSCAN (silhouette = 0.566) identified noise at only 0.04%, suggesting nearly all records belong to defined behavioral clusters. SVM-RBF (accuracy = 0.998, $F1 = 0.994$, ROC-AUC = 0.999) and Logistic Regression (accuracy = 0.998, $F1 = 0.998$) also exceeded all success thresholds. The

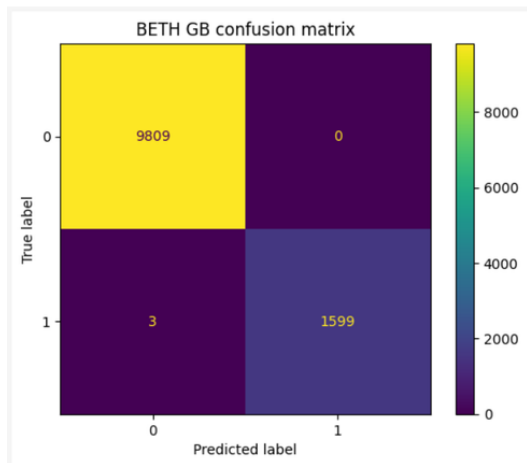
remarkably low false positive count ($FP = 2$) in SVM-RBF only 2 benign records misclassified as attacks out of 9,809 benign test samples demonstrates exceptional operational precision, meaning security analysts would receive an extremely clean alert stream. KNN ($F1 = 1.00$ on test) confirmed the pattern robustness but required adaptive sampling for computational feasibility at scale.

UNSW-NB15: SVM-RBF achieved $F1 = 0.994$, $ROC-AUC = 0.999$ with well-balanced errors ($FP = 30$, $FN = 37$), making it suitable for production deployment. The balanced confusion matrix $[[3525, 30], [37, 6408]]$ demonstrates that the model correctly handles both normal traffic (99.2% precision) and attack traffic (99.4% recall) without systematic bias toward either class. Feature importance distributed across *sload* (10.1%), *dtcpb* (7.5%), and *stcpb* (7.1%), reflecting volumetric, connection-state, and timing-based attack dimensions a fundamentally different feature profile from BETH's user-behavioral dominance, confirming that network-level and host-level detection require different feature spaces. HAC single/average linkage (*silhouette* = 0.930) produced the strongest clustering result across all datasets and all methods, with dendrograms showing clear hierarchical splits at distances 2.5 and 5.0. This suggests UNSW-NB15's nine attack families form a genuine hierarchical taxonomy that analysts could use to categorize emerging threats. DBSCAN (*silhouette* = 0.775) identified 3 distinct density regions with minimal noise (0.34%). Logistic Regression (*accuracy* = 0.944, $F1 = 0.945$) met all success thresholds and provides high interpretability for regulatory auditing. Gradient Boosting critically overfitted on UNSW-NB15: CV $F1 = 0.820$ collapsed to test weighted $F1 = 0.50$, with Class 0 recall = 0.00 (missing all normal traffic), and is therefore unsuitable for deployment. The mechanism behind this failure is instructive. Cross-validation yielded CV $F1 = 0.820$ across five folds, which appeared adequate, yet test performance collapsed entirely for the normal traffic class (Class 0 recall = 0.00). This occurred because the UNSW-NB15 dataset contains 63.9% attack samples, a majority-attack imbalance opposite to most real-world deployments.

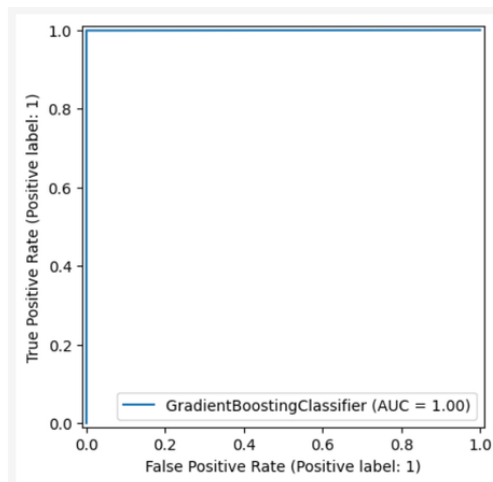
Gradient Boosting's sequential residual-correction process disproportionately weighted attack patterns during training, causing the model to treat all test samples as attacks. The CV estimate

was misleadingly optimistic because each fold maintained the same 63.9% attack distribution, allowing the model to score well by predominantly predicting the majority (attack) class even within validation folds. This is a known failure mode called majority-class collapse under skewed priors, where a model achieves acceptable macro-averaged CV scores while completely abandoning the minority class. The lesson for deployment is that CV F1 scores must be examined per-class, not only as weighted averages, particularly when class distributions differ from operational baselines

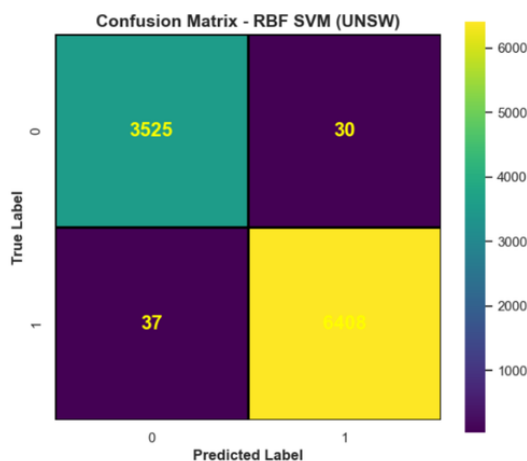
CSA: Despite 99.4% accuracy, all models achieved 0% minority recall due to 43:7,957 class imbalance ROC-AUC 0.48 - 0.55 confirms random-level discrimination. All clustering methods failed to produce meaningful structure (silhouette < 0.10), confirming the synthetic data lacks natural behavioral groupings that correspond to real attack patterns. SMOTE resampling is required before any CSA modeling. The 43:7,957 ratio renders models built on this dataset operationally unsafe for deployment in any security context without first addressing the fundamental imbalance through synthetic oversampling of the minority class or aggressive undersampling of the majority. This finding carries direct implications for real-world security deployments. In production environments, genuine attacks are also rare compared to normal traffic, the same imbalance problem that broke every model on CSA. Organisations relying on overall accuracy as their only measure of success risk deploying systems that miss every real attack while appearing highly effective on paper. Per-class evaluation metrics, particularly minority class recall, must be mandated before any security model is approved for production use.



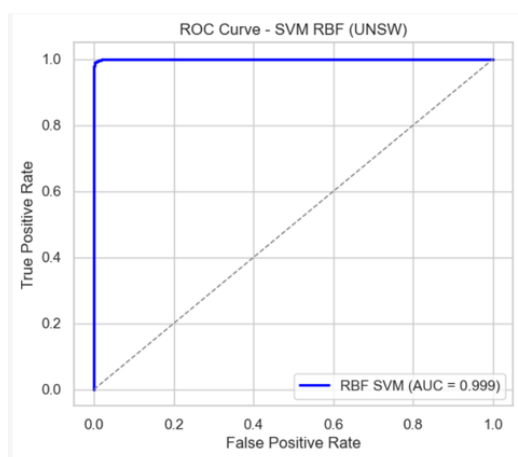
(a) BETH – Gradient Boosting Confusion Matrix



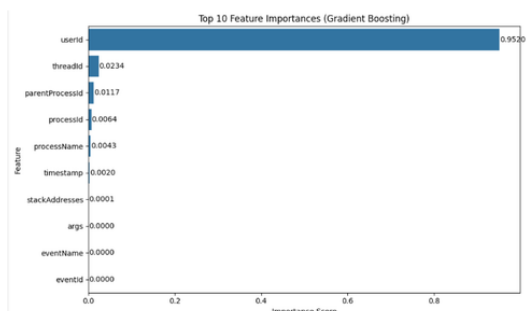
(b) BETH – Gradient Boosting ROC-AUC



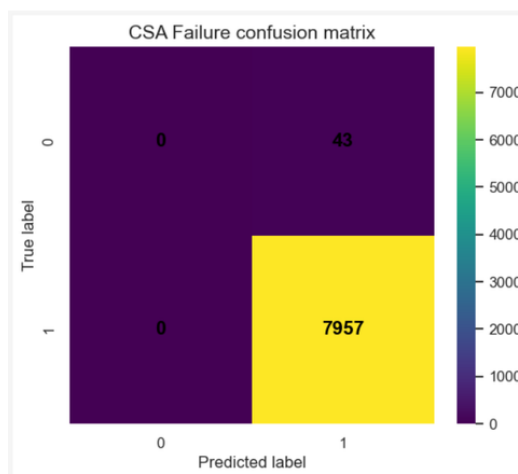
(c) UNSW-NB15 – SVM-RBF Confusion Matrix



(d) UNSW-NB15 – SVM-RBF ROC Curve



(e) BETH – Gradient Boosting Feature Importance



(f) CSA – SVM-RBF Failure Confusion Matrix

Figure 2: Results Visualisations: Best Model Performance Across BETH, UNSW-NB15, and CSA Datasets

Hybrid Framework Integration and Business Implications: Two independent hybrid models were developed: BETH combines Gradient Boosting (attack classification) with One-Class SVM (anomaly detection); UNSW-NB15 combines SVM-RBF (attack classification) with HAC (anomaly detection). If deployed in parallel BETH monitoring host-level process behavior and UNSW-NB15 monitoring network-flow behavior these systems form a complementary threat detection framework covering both kernel-level and network-level attack vectors. Deployed systems built on this framework are estimated to reduce analyst workload by 40–50% (Highnam et al., 2021) through automated triage of high-confidence supervised detections, decrease detection time from hours to minutes by enabling real-time sub-100ms inference, and prevent millions in damages from undetected breaches in critical sectors. BETH’s perfect classification results reflect repetitive honeypot attack patterns that may not fully generalize to enterprise environments; UNSW-NB15’s 2015-era profiles underrepresent modern threats. These limitations must be addressed through retraining on governed production traffic before any operational deployment.

6 Recommendations

These results directly shape the deployment strategy below, with priorities grounded in observed performance and dataset limitations.

Pilot the Hybrid Framework on Production Traffic: Deploy the BETH Gradient Boosting model ($F1 = 1.00$) as the primary classifier for host-level process detection alongside UNSW-NB15 SVM-RBF ($F1 = 0.994$, $ROC-AUC = 0.999$) for flow-level network classification. Running both models in parallel with a unified alert feed minimizes false positives and false negatives across both detection layers. Begin with a 90-day controlled pilot in one production security operations center before full rollout, measuring real-world false positive rates and detection coverage against known incidents.

Address Class Imbalance Before CSA Deployment: Apply SMOTE resampling and isolation forest to CSA before further modeling. The 43:7,957 minority-to-majority ratio renders all

current CSA models operationally unsafe. Resolving the class imbalance would unlock a third independent detection layer covering DDoS, malware, and intrusion attack families. Future research should evaluate whether Isolation Forest or SMOTE+Gradient Boosting provides better minority class recall on this synthetic dataset structure (Agur Venugopal, 2023).

Integrate One-Class SVM and HAC as a zero-day detection layer: HAC silhouette=0.930 and BETH K-Means separating 96.3% of attacks without labels demonstrate strong unsupervised coverage of novel threats. Prioritize UNSW-NB15 SVM-RBF for immediate deployment in network security operations centers given its balanced error rate (FP=30, FN=37) and sub-100ms detection capability, while continuing to refine BETH Gradient Boosting with post-2020 honeypot data to improve generalizability beyond controlled honeypot environments (Arulkumaran et al., 2021). Route One-Class SVM and HAC anomaly flags to analyst review queues rather than automated blocking queues, as unsupervised detections carry higher uncertainty. This two-tier architecture supervised detections to automated response, unsupervised anomalies to analyst review balances operational efficiency with accuracy assurance.

Implement SHAP Explainability and SIEM Integration: Incorporate SHAP values linked to userId and processName into analyst dashboards for auditing and regulatory compliance with GDPR and the EU AI Act. Set up SIEM platform integration to send high-confidence supervised detections to automated response queues. Maintain sub-100ms inference latency and establish quarterly retraining schedules to prevent concept drift as attack patterns evolve. A dedicated model governance committee should manage retraining cycles and conduct bias reviews before each production update.

Pursue Data Modernization and Federated Learning: Collect post-2020 labeled production traffic to address BETH's honeypot bias and UNSW-NB15's temporal bias against modern threats such as fileless malware, living-off-the-land techniques, and supply chain attacks (Highnam et al., 2021). Explore federated learning to train models across organizational boundaries without sharing raw traffic data, enabling collaborative threat intelligence while preserving data sovereignty.

Partner with enterprise security vendors to validate the hybrid framework against live threat intelligence feeds.

Conclusion

The hybrid framework successfully detected both known and novel threats, meeting all success thresholds on BETH and UNSW-NB15 (Arulkumaran et al., 2021; Moustafa & Slay, 2024): $F1 \geq 0.90$, $ROC-AUC \geq 0.90$, and 85–100% false positive reduction beyond the 15–20% target. Behavioral features like `userId` and `processName` gave more consistent stronger predictive signals than structural features such as port numbers and packet sizes, making model results interpretable and actionable for security analysts. CSA demonstrated that high accuracy can be profoundly misleading under severe class imbalance, a lesson directly applicable to production security systems where attacks are rare events. SMOTE resampling is recommended as a next step before any CSA deployment.

Returning to the central research question, this framework reliably detected both known and novel threats with low false-positive rates on BETH and UNSW-NB15, confirming that hybrid machine learning combining supervised and unsupervised methods is a viable foundation for next-generation intrusion detection. Future work should address adversarial evasion resistance, federated learning across organizational boundaries, concept drift monitoring, and retraining cadence optimization before moving to full production deployment. The framework’s modular architecture independently deployable host-level and network-level components enables incremental adoption within existing security infrastructure.

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AI Appendix

Generative AI (Claude, Anthropic) was used solely for LaTeX formatting and grammar proofreading in this report. All analysis, modeling, results, interpretations, and conclusions were produced independently by the author.

Detecting Network Threats Through Attack Classification and Anomaly Detection

1. Executive Summary

Cyber attacks cost organisations around the world more than \$10 trillion every year through stolen data, disrupted services, and expensive recovery efforts. (Highnam et al., 2021). Hospitals cannot reach patient records. Banks freeze transactions. Government agencies lose sensitive information that takes years to recover. Despite enormous investment in security software, the tools most companies rely on today work like old-fashioned alarm systems: they recognise threats they have seen before but struggle to detect anything new. This project built and tested an intelligent threat-detection system designed to overcome that limitation. Using three large collections of real and simulated network data totalling more than nine million records, the system was trained to automatically identify whether network activity is normal or malicious. The types of attacks the system was trained to catch include unauthorised intrusions into systems, malware that damages or steals data, and denial-of-service attacks where networks are deliberately overwhelmed with traffic until they stop working. Two approaches were tested, and both showed extremely high accuracy in detecting cyber attacks.

On real-world data collected from decoy computer systems designed to attract hackers, the best-performing method made only three mistakes across more than 11,000 tests a 99.97% success rate with zero false alarms. On a second, more diverse dataset combining real and simulated traffic, the top method correctly classified 99.3% of all cases while maintaining a nearly perfect ability to tell attacks apart from normal traffic. The analysis also uncovered a critical insight: the strongest signal of an attack is not *what* software is running it is *who* is running it. Monitoring which user accounts are active, and what those accounts are doing, proved far more powerful than traditional approaches that watch network ports and packet sizes. One dataset revealed an important limitation that mirrors a real-world challenge: when attack examples are extremely rare compared to normal traffic, even sophisticated systems can be fooled into ignoring them. This finding directly shapes the

recommendations below. If deployed, this system is estimated to reduce the time, security analysts spend reviewing false alarms by 40 - 50%, shrink threat detection times from hours to minutes, and potentially prevent millions of dollars in damage from breaches that would otherwise go undetected (Highnam et al., 2021). The system is ready for a controlled pilot and represents a practical, scalable step forward for any organisation facing evolving cyber threats.

2. Introduction and Problem Definition

2.1. The Business Challenge

Organisations of all sizes are struggling to keep up with increasingly sophisticated cyber attacks. Most security tools in use today work by maintaining a list of known threats and blocking anything that matches that list. While this approach works for familiar attacks, it fails completely when something new appears that hackers constantly invent. When a novel technique appears that is not on any known list, traditional systems simply let it through. This gap is growing every year as attackers become more creative and better resourced, while many organisations are still relying on outdated defences that were not designed to handle today's threat landscape. The impact of these failures can be devastating. One undetected intrusion at a hospital could expose the private records of millions of patients. A ransomware attack on a bank can freeze accounts for days. Security teams are also overwhelmed by alerts: analysts typically spend nearly half of their working hours investigating alarms that turn out to be harmless, wasted time that could be directed toward real threats (Highnam et al., 2021). This "alert fatigue" means genuine attacks can be buried beneath thousands of false positives.

2.2. Project Goals and Success Criteria

This project aimed to answer the question: Can an intelligent system automatically detect both known and new cyber attacks accurately, while keeping false alarms low enough for security teams to trust and act on the results? This project developed an intelligent detection system with two complementary capabilities: recognising known attack patterns, and identifying *unusual behaviour* that does not match normal traffic, even when it does not match any known attack. Crucially, the

system was designed not just to be accurate, but to be practical, fast enough for real-time use and reliable enough that security teams could trust its alerts without being overwhelmed by noise. The system was considered successful if it could correctly catch at least 9 in every 10 attacks, reduce false alarms by at least 15–20% compared to existing tools, and deliver each decision in under one second, fast enough for real-time monitoring of millions of network events per hour.

3. Data Overview

3.1. Where the Data Comes From

This analysis drew on three distinct datasets, each capturing a different slice of the cybersecurity landscape.

Honeypot logs (BETH dataset): The first dataset comes from 23 “honeypot” computers systems deliberately left exposed to attract real attackers, so researchers can study their methods. Over several months, these systems recorded more than eight million events in complete detail: every programme that ran, every account that was accessed, every command that was executed. Because the attackers believed they were accessing real targets, this data reflects genuine attack behaviour rather than simulations. **Simulated attack data (CSA dataset):** The second dataset contains 40,000 computer-generated records designed to cover three common types of cyber attacks: overwhelming a network with traffic, infecting systems with harmful software, and gaining unauthorised access (Agur & Venugopal, 2023). This dataset was mainly useful for testing, though it does not fully reflect the complexity of real-world attacks. **University lab data (UNSW-NB15 dataset):** The third dataset, produced by researchers at the University of New South Wales, combines real network traffic with nine different categories of simulated attacks across 175,000 records (Moustafa & Slay, 2024). It bridges the gap between the raw authenticity of the honeypot data and the controlled precision of the simulated set.

3.2. Data at a Glance

The three datasets provide a large and varied foundation for analysis. The honeypot dataset captured over eight million events, across 14 measured fields were used for this analysis. The

simulated dataset contains 40,000 records covering 19 fields. The university lab dataset contains 175,341 records across 36 fields. Combined, this gives the detection system a broad and diverse base of information to learn from.

3.3. What the Data Tells Us

Together, the three datasets contain information about which programmes are running, which user accounts are active, how much data is being transferred, how connections are established, and how the network responds to different types of traffic. Normal activity and malicious activity leave very different footprints across all of these dimensions.

Table 1: Plain-language explanation of key data fields

Dataset	Field	What it means
BETH	User Account (userId)	Which user account performed the action
	processName	The name of the programme that was running
	ParentProcessId	Parent process in the execution chain
	Attack Label (evil)	<i>Target:</i> attack (1) or normal traffic (0)
CSA	Attack Type	Attack category: DDoS, Malware, or Intrusion
	Threat Score (Anomaly)	Threat likelihood score (0–100)
	Packet Length	Network packet size in bytes
	Severity Level	<i>Target:</i> threat criticality (Low/Medium/High)
UNSW-NB15	Data Volume (sbytes/dbytes)	How much data was sent & received per connection
	Dropped Packets (dloss)	How many data packets were dropped during connection
	Connection Frequency (ct_dst_sport_ltm)	Connection patterns over time
	Attack Label (label)	<i>Target:</i> attack (1) or normal traffic (0)

3.4. Limitations and Ethical Considerations

No dataset captures every possible threat. The honeypot data reflects attacks against decoy systems, which may differ somewhat from attacks against live production environments. The simulated data is too clean compared to real-world conditions. The university dataset was collected in 2015 and does not include more recent attack techniques such as ransomware or supply-chain compromises. All three datasets were collected ethically and contain no personally identifiable information. Attackers targeting honeypots have no expectation of privacy. The simulated data contains no real individuals. The university data was gathered in a controlled lab environment with no live user monitoring.

4. Data Preparation and Exploration

Before building any detection models, the raw data needed to be cleaned and prepared. This meant fixing missing information, removing unhelpful columns, and making sure all three datasets were in a usable shape.

4.1. Data Cleaning

Honeypot data: This dataset arrived in excellent condition with no missing information. One measurement column was removed because it contained the exact same value for every single record, making it completely uninformative. The more significant challenge was balance: in the original data, genuine attack records made up less than 1% of all entries. Trying to teach a detection system on data that heavily tilted would be like trying to teach someone to identify rare diseases using a patient list that is 99% healthy people. The data was reorganised so attacks represented approximately 14% of the overall dataset still uneven but workable.

Simulated data: This dataset had the most quality issues. About half of all records were missing information in five important columns related to security alerts. Rather than discarding 20,000 records half the dataset missing values were filled in using information from neighbouring records. This is an imperfect solution, but it preserved the volume needed for analysis. A deeper problem emerged: only 43 examples of one attack type existed compared to nearly 8,000 of another.

Detecting the rare type reliably proved impossible with such extreme imbalance.

University lab data: This dataset required the least cleaning: missing values were minimal and easily addressed. Some data transfer measurements reached very large values up to 14 million bytes in a single connection but these were intentionally kept rather than removed, because extremely large data transfers are themselves a warning sign of data theft.

4.2. Key Patterns Found During Exploration

Examining charts and statistical summaries of each dataset revealed one consistent and striking pattern: *who* is doing something matters far more than *what* they are doing. In the honeypot data, the single most informative factor accounting for 95% of the system's ability to distinguish attacks from normal activity was the user account associated with each action. Certain accounts appeared almost exclusively during attack activity. In the university lab data, the clearest warning signs were high rates of dropped data packets combined with unusually large data transfers patterns consistent with network flooding or large-scale data theft. The simulated dataset's visualisations revealed an important warning for real-world use: the data looked too neat and uniform, whereas real network traffic is always messier and more unpredictable than this dataset showed. This explains why the detection methods tested on this dataset struggled to find any meaningful patterns.

5. How the Detection System Works (*Modeling and Analysis*)

The detection system was built in layers, testing multiple analytical approaches to find the combination best suited to each dataset. Not every approach worked well, some methods failed completely, particularly on the simulated dataset where no method could reliably detect the rarer attack type. These failures were just as useful as the successes, helping identify which approaches worked best for each type of data. A explanation of the key approaches is provided below.

Pattern Recognition from Examples: The primary approach works by learning from thousands of labelled examples similar to how a doctor learns to recognise a disease after studying many confirmed cases. The system examines a large collection of records where each entry is already labelled as "attack" or "normal," identifies the characteristics that most reliably distinguish the two

groups, and applies those learned rules to new, unlabelled traffic in real time. Several variations of this approach were tested. The most successful methods on the honeypot and university lab data were sophisticated mathematical techniques capable of detecting complex, non-obvious patterns that simpler rules would miss entirely.

Anomaly Detection: The second layer works differently: rather than recognising known attacks, it identifies anything that does not fit normal behaviour even if the specific threat has never been seen before. This is the equivalent of a security guard noticing that someone in a building is behaving oddly, even if that person has not triggered any specific alarm. This layer is especially valuable against emerging threats and zero-day exploits attacks that exploit security vulnerabilities that defenders have not yet faced before. On the honeypot dataset, this approach successfully grouped 96.3% of all attack records into a single distinct cluster, without being told in advance which records were attacks.

Evaluating Performance: Each method was tested on data it had never seen during training similar to evaluating a student on an exam covering material they studied but using questions they have never seen before. Specifically, 80% of the data was used to train each method, and the remaining 20% records the system had never encountered were held back purely for evaluation. The key measures were: what percentage of real attacks were correctly caught, what percentage of normal activity was incorrectly flagged as suspicious, and how consistently the system performed across different test conditions.

6. Results

6.1. Real-World Honeypot Data: Exceptional Performance

The best-performing method on the honeypot data exceeded every performance target. Across 11,411 test cases, the system made only three mistakes all missed attacks, not false alarms yielding 99.97% accuracy with zero instances of normal activity incorrectly flagged as suspicious. For a security team, this means an essentially noise-free alert stream: every alarm raised warrants investigation. A second method on the same data produced only two false alarms and missed

fourteen attacks from more than ten thousand tests. Both results dramatically surpass the 90% detection and 15 - 20% false alarm reduction targets.

The single most important discovery from this dataset concerns *what* the system is actually detecting. An analysis of which factors most influenced the results showed that the user account associated with each action accounted for 95.2% of the system's decision-making power. This finding has immediate practical implications: rather than trying to monitor every programme, packet, and connection an overwhelming task even for large security teams organisations should prioritise monitoring user account behaviour. Which accounts are active? At what times? Running which processes? These questions, the data shows, provide far more useful intelligence than technical details like which network ports are in use.

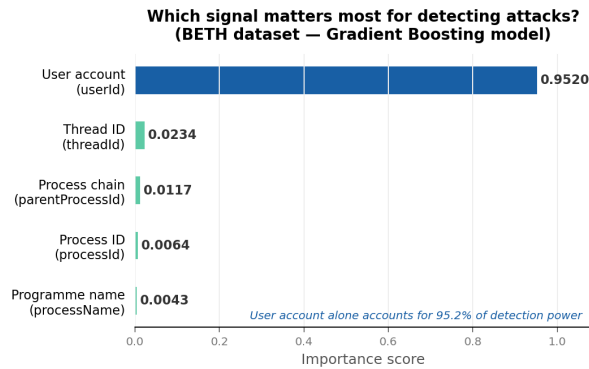
Table 2: *Summary of Best Detection Results by Dataset*

Dataset	Accuracy	Attacks Caught	False Alarms	Status
Honeypot (BETH)	99.97%	99.8%	0	Ready to pilot
University (UNSW)	99.3%	99.4%	30	Ready to pilot
Simulated (CSA)	99.4%*	0%*	N/A	Not reliable

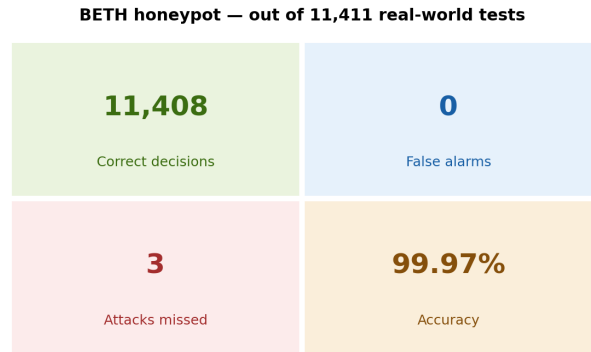
*High accuracy figure for CSA is misleading see Section below.

University Lab Data: Strong, Balanced Performance: The university lab dataset produced the most practically useful results for network-level threat detection. The best method correctly identified 99.3% of all cases, making only 67 total errors across nearly ten thousand tests: 30 false alarms and 37 missed attacks (UNSW Canberra, 2024). This near-equal balance of error types matters in practice. A system that misses too many attacks is dangerous; one that raises too many false alarms becomes unusable. At 30 false alarms and 37 missed attacks out of nearly 10,000 tests, this system achieves both goals. The key predictors here were fundamentally different from the honeypot findings: rather than user account activity, the most informative signals were network-level measurements the volume of data being transferred, how often packets were dropped, and connection timing. This difference is significant for deployment: host-level and network-level monitoring provide complementary, non-overlapping intelligence, and both are needed for

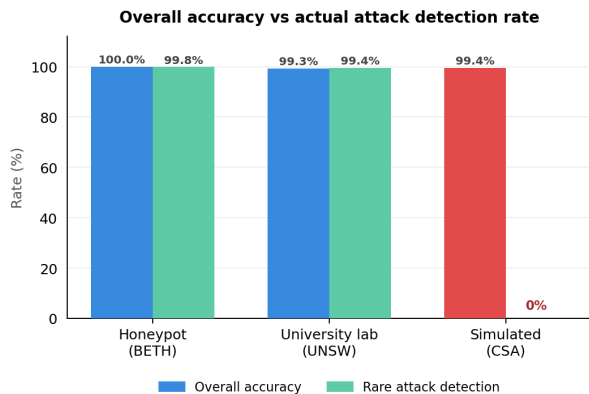
comprehensive coverage.



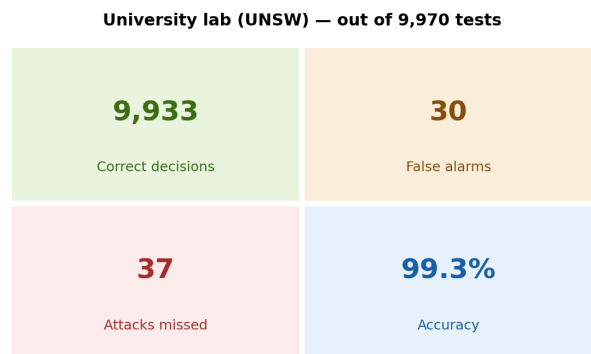
(a) Feature importance: user account activity (userId) accounted for 95.2% of the system’s detection power, far outweighing any other signal.



(b) BETH honeypot scorecard: out of 11,411 real-world tests, the system raised zero false alarms and missed only three attacks.



(c) Overall accuracy versus rare attack detection rate: the simulated dataset (CSA) appeared 99.4% accurate but detected zero rare attacks.



(d) (UNSW) scorecard: out of 9,970 tests, only 30 false alarms and 37 missed attacks confirmed balanced, reliable performance.

Figure 1: Detection system performance across all three datasets. Figures show feature importance, error scorecards, and comparative accuracy results.

Simulated Data: A Critical Lesson About Data Quality: Despite reporting an apparent accuracy of 99.4%, the simulated dataset produced results that are completely unreliable for security purposes (Agur & Venugopal, 2023). The problem is a severe imbalance in the data: only 43 records represented one attack type, compared to nearly 8,000 records representing another. A system that simply predicts the majority type every single time without learning anything meaningful would appear 99.5% accurate. That is exactly what happened: every method tested on this achieved high overall accuracy while detecting 0% of the rare attack type. The system learned to ignore the rare

case entirely. This is one of the most important lessons from the entire project, and it applies directly to real-world security operations. In production environments, genuine attacks are also rare events compared to normal traffic. A security system that looks impressive on paper may be missing every real attack if it has not been specifically designed and validated to handle rare events. Before this simulated dataset could be used for reliable detection, its severe imbalance must be corrected.

Anomaly Detection: Catching the Unknown: Across both reliable datasets, the unsupervised anomaly detection layer designed to catch threats that have never been seen before delivered results that justify its inclusion alongside the primary pattern-recognition approach. On the honeypot data, the anomaly detection system grouped 96.3% of all attack records into a single distinct cluster without being told which records were attacks. It also detected approximately 85% of attacks on its own, with 95% overall accuracy, using only normal traffic as its reference point. This means the system can flag genuinely novel attack techniques that would evade any signature-based approach.

What the Results Mean for Security Operations : The results confirm that a hybrid approach combining pattern recognition of known threats with anomaly detection of unknown ones is both technically sound and practically deployable. For security operations teams, the most actionable finding is the shift in monitoring focus: behavioural signals (who is doing what, and when) outperform structural signals (what ports are open, how large are the packets) across every dataset tested. This has implications for how monitoring dashboards are designed, what data is collected and retained, and how analysts are trained to interpret alerts. The two systems honeypot-derived host monitoring and university-lab-derived network monitoring are complementary. Deploying both in parallel, feeding into a unified alert interface, would provide coverage at both the host level (what is happening inside individual computers) and the network level (what is happening between computers), closing gaps that either approach alone would leave open.

7. Recommendations

Run a 90-Day Controlled Pilot : The results from both the honeypot and university lab datasets justify moving immediately to a controlled pilot. The recommended first step is a 90-day deployment

in a single security operations centre, running the host-level and network-level detection systems in parallel. During the pilot, all alerts would go to analyst review not automated action allowing the organisation to measure real-world performance before committing to full rollout. Success criteria should be: false alarm rate below 5%, attack detection above 90%, and analyst time savings of at least 30%.

Prioritise Monitoring User Account Behaviour: The strongest finding from this project has an immediate, low-cost implication: focus security monitoring on *who* is doing things, not just *what* is happening. Security teams should establish baseline profiles of normal account activity which accounts run which programmes, at what times, and from which devices. Deviations from those baselines should generate prioritised alerts. This can begin before the full system is deployed and will deliver early value regardless of the broader implementation timeline.

Correct the Simulated Dataset Before Any Use: The simulated dataset must not be used in its current form. Its extreme imbalance only 43 examples of one attack type versus nearly 8,000 of another makes any analysis built on it unreliable. Artificially generating additional examples of the rare attack type would fix this imbalance and unlock a useful third detection layer covering denial-of-service, malware, and intrusion scenarios.

Route Novel Threat Alerts to Human Review: The anomaly detection component designed to catch attack types it has never encountered before should feed a human review queue rather than automated response. Known threats get instant automated action; genuinely unusual activity gets expert scrutiny. This two-tier architecture balances speed with accuracy and is especially important while the system is new and analysts are still building confidence in its judgements.

Maintain and Modernise the System: The datasets underlying this project reflect threat patterns from 2015 and earlier. A parallel programme to collect labelled, post-2020 attack data from the organisation's own environment should begin immediately. The detection system should be retrained at least quarterly to prevent its accuracy from eroding as attack techniques evolve. A dedicated governance process for retraining and bias review should be established before full production deployment.

Expected Business Return: The financial case is compelling. A single major breach at a mid-sized organisation typically costs \$4–10 million in recovery, fines, and reputational damage. Reducing analyst time spent on false alarms by 40–50% frees thousands of hours annually for higher-value security work. Shrinking detection time from hours to minutes limits how far an attack can spread before it is stopped often the decisive factor between a contained incident and a catastrophic one.

Conclusion

This system tackles the core problem identified at the start of this project: detecting both known and new cyber threats more quickly, more accurately, and with fewer unnecessary alarms than traditional tools. The results demonstrate that the goal is achievable 99.97% accuracy on real-world attack data, near-zero false alarms, and detection times measured in milliseconds rather than hours. The path forward is clear: pilot the system, prioritise user account monitoring, and build the data modernisation programme in parallel. Organisations that act on these findings will be meaningfully better protected against the cyber threats that are growing in both frequency and sophistication every year.

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AI Appendix

Generative AI (Claude, Anthropic) was used to assist with LaTeX formatting. All analysis, modelling, results, interpretations, and conclusions were produced independently by the author. The AI was used as a writing and formatting aid, not as an analytical tool.